INCH-POUND

MIL-C-3098/66G
28 June 1995
SUPERSEDING
MIL-C-3098/66F
8 March 1983

MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR88/U

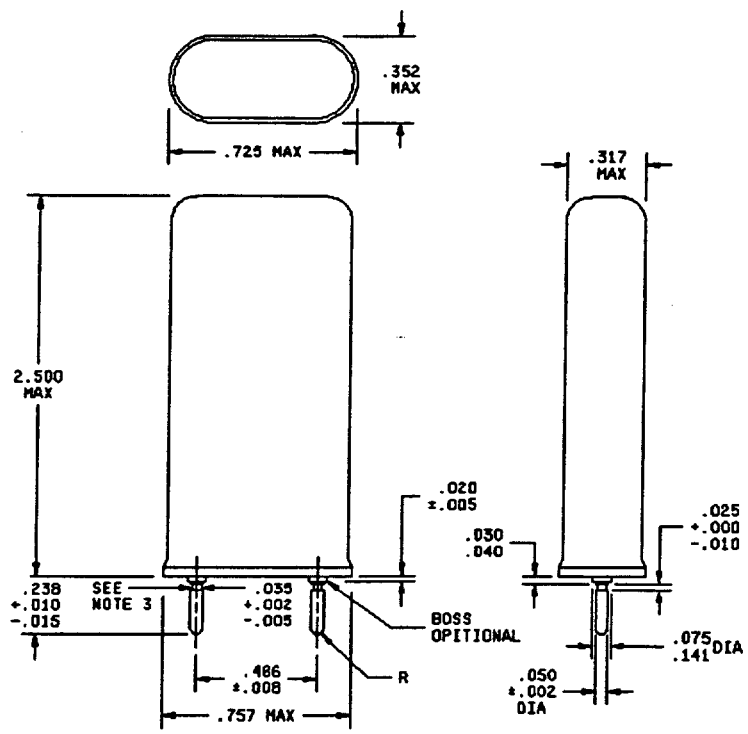
INACTIVE FOR NEW DESIGN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for the acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 2.75 to 16 kHz; fundamental; noncontrolled; series resonance.

Inches	mm
.002	0.05
.005	0.13
.008	0.20
.010	0.25
.015	0.38
.020	0.51
.025	0.64
.030	0.76
.035	0.89
.040	1.02
.050	1.27
.075	1.90
.141	3.58
.238	6.05
.317	8.05
.352	8.94
.486	12.34
.725	18.42
.757	19.23
2.500	63.50



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Pin undercut may be omitted.
4. Marking to be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR88/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: In accordance with figure 1 and MIL-H-10056/4 (HC-13/U, modified). 1/ 2/

Dimensions, marking, and configuration: See figure 1.

Frequency range: 2.75 to 16 kHz, inclusive.

Frequency tolerances:

Primary operating temperature range: ±50 ppm.

Secondary operating temperature range: ±100 ppm.

Equivalent resistance: 100,000 ohms, maximum.

Mode of oscillation: Fundamental.

Operating temperature ranges:

Primary: 0°C to +65°C, inclusive.

Secondary: -40°C to 0°C and +65°C to +70°C, inclusive.

Rated drive level: 0.1 mW.

Capacitance, shunt:

kHz	pF, maximum
2.75 to 8.0	15
8.0+ to 16.0	30

Shock (specified pulse):

Frequency change permitted: 10 ppm.

Equivalent resistance change permitted: ±15 percent.

Thermal shock:

Frequency change permitted: 10 ppm.

Equivalent resistance change permitted: ±15 percent.

Vibration: Method 201 of MIL-STD-202, amplitude .015 inch (0.38 mm) (.030 inch (0.76 mm) total excursion).

Frequency change permitted: 10 ppm.

Equivalent resistance change permitted: ±15 percent.

Bond strength: 800 grams, minimum.

1/ The socket designed to mate with crystal holder HC-13/U is Part or Identifying Number (PIN) M12883/04-01 (see MIL-S-12883/4).
For Navy use, other sockets listed in MIL-STD-242 may be used.

2/ Crystal holder (see figure 1) is similar to type HC-13/U of MIL-H-10056/4. "Holder, Crystal, Type HC-13/U," except for cover height dimension.

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QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

NOTE: This crystal-unit type is to be used for special applications and for replacement purposes only. Specific approval of the Government is required prior to each intended use. See MIL-STD-683.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SM,
Navy - AS, MC, SH
Air Force - 17, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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